Latha Mathavan Engineering College

IIEC INNOVATION INCUBATION
ENTERPRENURSHIP CELL

Final year CSE

Latha Mathavan
Group of Institutions

GluGuard Project

TEAM MEMBERS



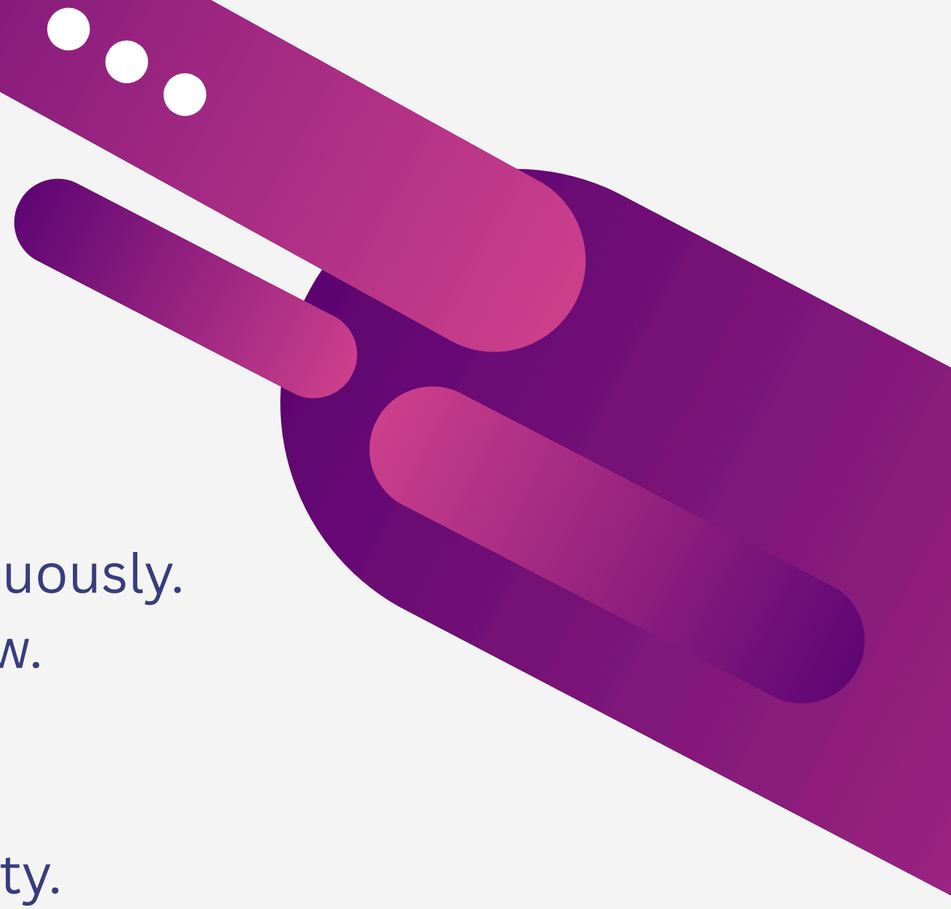
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Smart monitoring saves lives before danger begins

INTRODUCTION



- GluGuard is a smart glucose drip monitoring system for hospitals.
- It monitors the glucose level in the IV drip bottle continuously.
- The system detects when the glucose level becomes low.
- It alerts nurses or caretakers using a buzzer and LED.
- This reduces the need for frequent manual checking.
- It helps prevent the glucose bottle from becoming empty.
- The system improves patient safety and reduces human errors.
- GluGuard is simple, reliable, and cost-effective for healthcare use.



Problem Statement

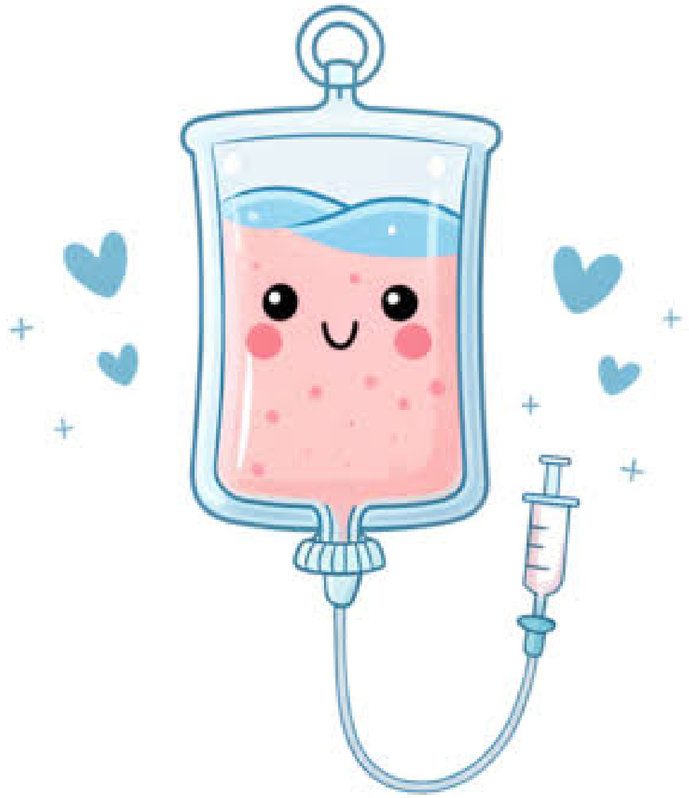
In traditional hospital setups, nurses must frequently check glucose bottles for multiple patients. This creates several problems:

- It increases workload for nurses
- There is a chance of delay in checking
- If the glucose bottle becomes empty, air may enter the patient's vein, which is dangerous
- Continuous monitoring is not possible manually

Hence, there is a need for an automated system that can monitor glucose levels accurately and provide alerts on time.

Objectives

To develop a smart, low-cost, IoT-based IV glucose monitoring system that continuously monitors the glucose level, alerts healthcare staff when the bottle is nearly empty, and automatically stops the drip to improve patient safety.



Purpose

- To develop an automated glucose drip monitoring system.
- To continuously monitor the glucose level in the IV drip bottle.
- To reduce the need for frequent manual checking by nurses.
- To provide alerts using a buzzer and LED when the glucose level is low.
- To prevent the glucose bottle from becoming empty and reduce the risk of air entering the patient's vein.
- To improve patient safety and healthcare efficiency.
- To reduce the workload of nurses and caretakers.
- To provide a simple, reliable, and cost-effective monitoring solution.

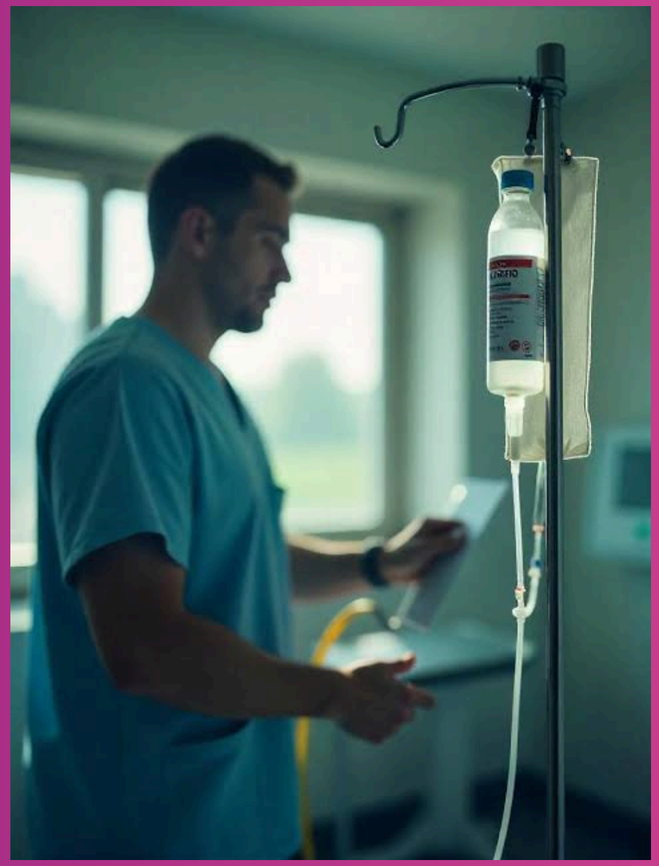
Existing System

Overview

In the existing healthcare system, IV glucose bottles are monitored manually by nurses or hospital staff. They regularly check the glucose level and replace the bottle when it becomes empty. This process depends entirely on human observation and timely intervention.

How It Works

1. The nurse hangs the IV glucose bottle.
2. The glucose flows into the patient's vein through the IV set.
3. Nurses periodically check the liquid level.
4. When the bottle is nearly empty, they manually replace it.
5. The drip continues after the new bottle is connected.



Limitations of Existing System

- Human Error
- Time Delay
- Continuous Manual Monitoring
- Increased Workload
- Risk of Air Entry in IV Line
- No Automatic Alert System
- No Automatic Flow Control
- Patient Safety Risks

Advantages

Reduces Manual Work

Nurses do not need to check glucose bottles frequently, saving time and effort.

Improves Patient Safety

Prevents risks such as air entering veins when glucose bottle becomes empty.

Low Cost

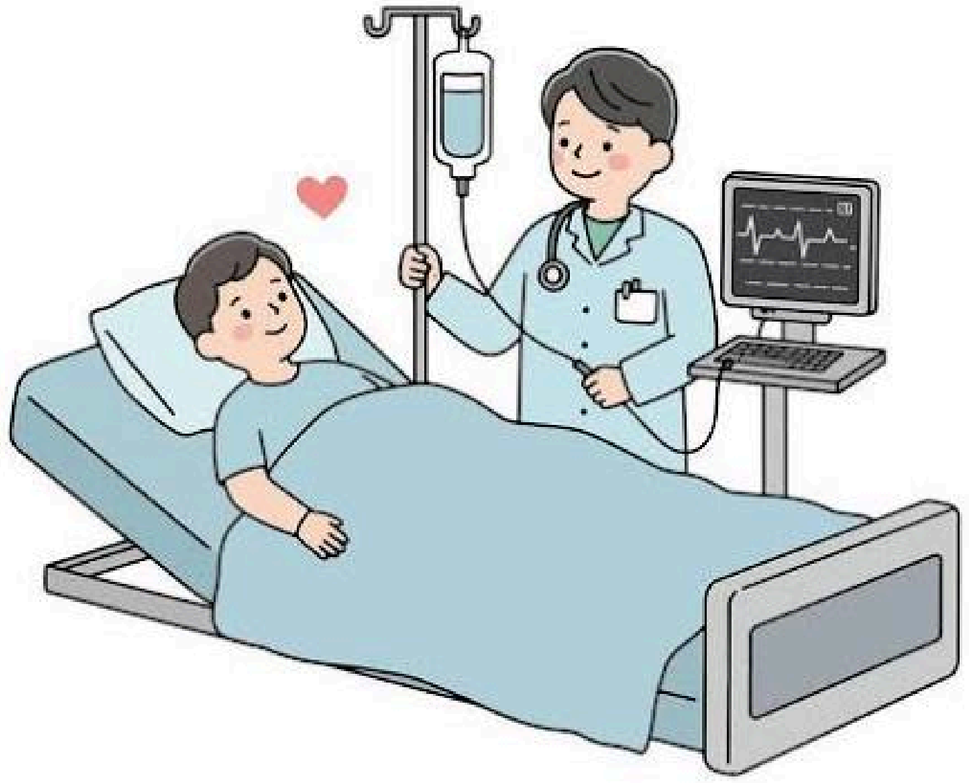
Uses simple and affordable components like Arduino and sensors.

Continuous Monitoring

Provides real-time tracking without interruption.

Easy to Use

Simple setup and operation make it suitable for hospitals and home care

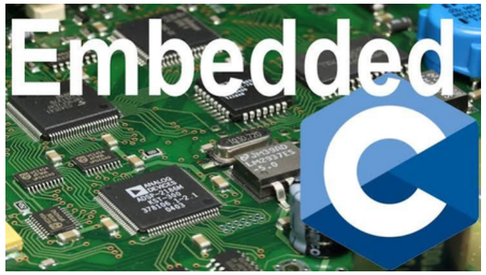


Hardware Components

- Arduino UNO – Main Controller
- Capacitive Liquid Level Sensor – Detects IV fluid level
- Buzzer – Audio alert
- LED – Visual indication
- Breadboard & Jumper Wires – Circuit connections
- Power Supply – Provides electrical power

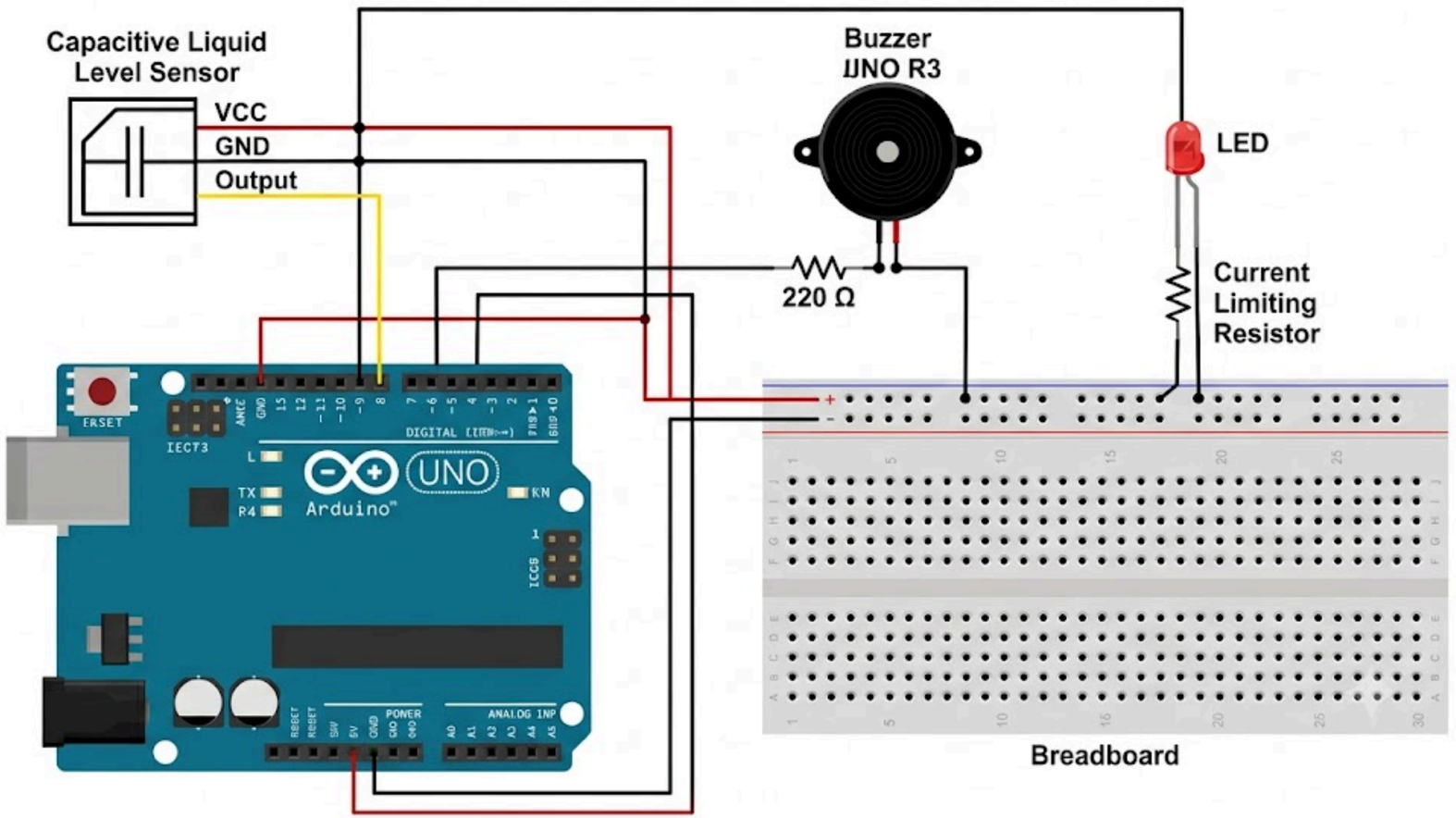
Software Requirements

- Arduino IDE
- Embedded C Programming
- USB Driver
- Windows Operating System





Circuit Diagram





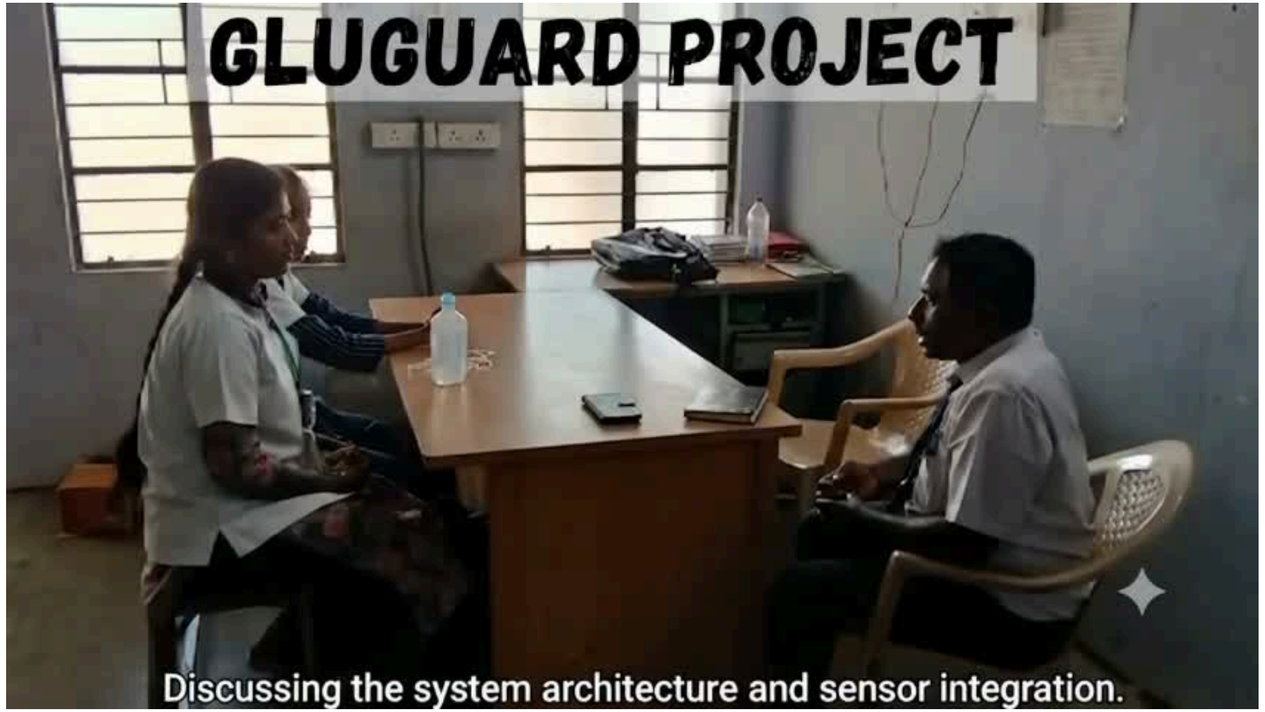
Basic Prototype

he prototype of this project is currently under development and will be completed soon. We are working on integrating the hardware components with advanced technologies to build a fully functional system.

Prototype design is in progress

Progress Status





System Overview

The GluGuard system works by measuring the weight of the glucose bottle using a load cell sensor. As the glucose flows into the patient, the weight of the bottle gradually decreases.
The system continuously tracks this weight change and calculates the remaining glucose level. The data is displayed in real-time on an LCD screen, and alerts are generated when the level goes below a threshold.



Applications

- Hospitals
- Clinics
- Intensive Care Units
- Home Healthcare
- Emergency Ambulances
- Rural Health Centers





Key Features

- Automatic glucose monitoring
- Real-time detection
- Non-contact sensing
- Automatic drip control
- Audio and visual alerts
- Low-cost solution
- Easy installation
- Safe operation



GluGuard Advance Model

Automatic Switch-Off Mechanism
Automatically stops the glucose flow when the bottle reaches the preset minimum level.
Prevents air from entering the IV line after the bottle becomes empty.
Reduces the need for continuous manual monitoring by nurses.



Market Potential & Business Model

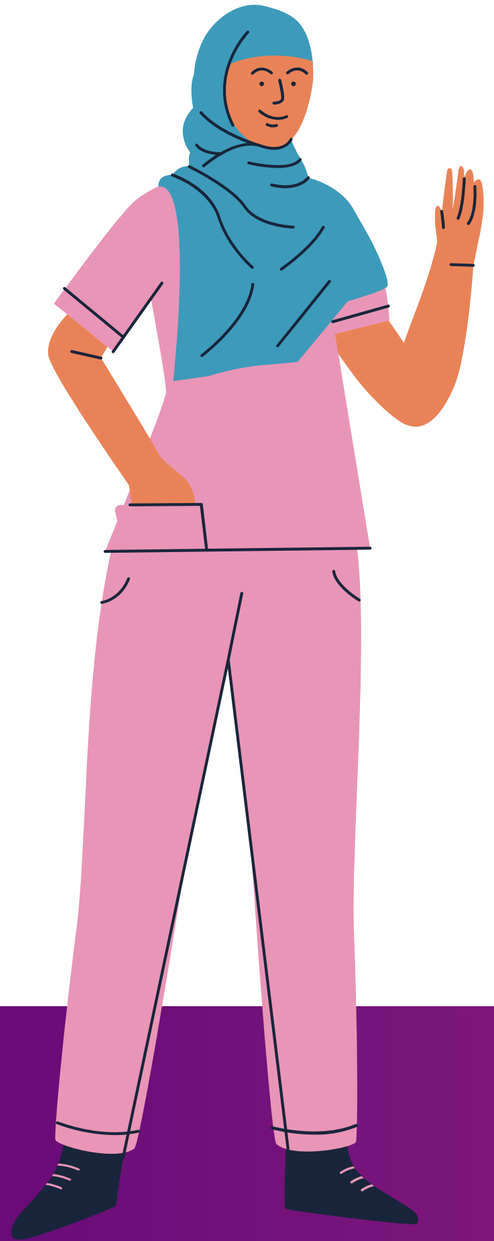
Target Customers

- Hospitals
- Clinics
- Nursing Homes
- Home Healthcare Providers

Revenue Model

- Device sales
- Annual maintenance contracts
- Hospital subscriptions
- IoT monitoring services





Conclusion

The Patient Health Monitoring System using IoT is an efficient and reliable solution for continuous health tracking. It reduces manual effort, improves patient safety, and enables real-time monitoring. This system is highly useful in modern healthcare environments.

Thank you!



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